Task 2: Data Understanding and Background

SINTEF Ocean Real Vessel Data Challenge | TDT4259 Applied Data Science

25 October 2026 | Preliminary technical findings

1. Dataset inventory

The dataset contains **41,079,968 sensor records in twelve 2024 monthly files** and a further **1,112,405 records for January 2025**, giving **42,192,373 records** across thirteen files. The sensor registration identifies 37 QIDs. Thirty-six occur in the 2024 data; 26 occur in January 2025. The registered channel without observed values is a turbocharger RPM indicator. All torquemeter channels, including the shaft-power target, are absent in January 2025, so that month is not a labelled test period.

The quality scan parsed all 42,192,373 observed sensor values as finite numeric values and found no malformed short records or duplicate original **(timestamp, QID)** keys. Recorded numeric completeness should not be confused with coverage across calendar time.

1.1 Record structure and types

Field	2024 sensor records	January 2025 records
Timestamp	ISO 8601 text with a `Z` suffix, interpreted as UTC-aware datetime	Day-first timestamp text without a timezone; its actual timezone is unknown
QID	Hierarchical text identifier; categorical key	Hierarchical text identifier; categorical key
Sensor value	Numeric text parsed as `float64`	Numeric text parsed as `float64`
Number of Records	Not present	Additional integer-formatted field; precise meaning unconfirmed

The dictionary classifies the **36 observed channels** as numeric (`float64` after parsing) and the unobserved registered channel as **not observed**. Direction and heading measurements are numeric but also have circular-angle semantics. The 2024 source records are not necessarily sorted chronologically; ordering must be handled explicitly during processing.

2. Sampling, temporal alignment and coverage

The different sensors operate at different rates and timestamps. A wide table formed from the union of every raw timestamp therefore contains numerous empty cells even when individual sensors recorded data correctly.

October 2024 target measure	Result	Meaning
Numeric values among recorded target rows	88,304 / 88,304	No null value inside a recorded target row
Target presence on the raw union-timestamp grid	88,304 / 376,474	About **76.5%** of grid cells blank, largely because timestamps differ
Target presence on completed-minute grid	22,088 / 44,640	**49.48%** of calendar minutes represented by a target observation

For the October descriptive analysis, onboard readings are aggregated over trailing **one-minute intervals**, each labelled by the end of the completed minute. Environmental values are associated with an interval only when their raw timestamp is no later than the interval end and their age is at most sixty minutes. This avoids using a later *raw timestamp* for an earlier interval, but does **not** prove when a provider actually issued or made that value available.

There are **14 external provider channels**. Each has 643 raw October readings, **38,601** aligned calendar-minute values (86.47%) and **22,088** minutes paired with the power target. The target-paired values derive from **369 distinct raw provider timestamps** per channel. Scavenging-air pressure is an onboard engine

measurement: it covers **22,092 completed minutes**, with **22,087** target-paired minutes. Water depth has only 61 completed-minute observations and no overlap with the October target.

2.1 Monthly observation-count density

The percentages below compare actual target counts with a *nominal* 15-second schedule across each calendar month. They indicate recording density, not the cause of gaps or verified sensor uptime.

Month	Recorded target readings	Nominal density	Negative readings
January	166,699	93.4%	115
February	166,389	99.6%	111
March	149,512	83.7%	30
April	171,160	99.1%	20
May	170,498	95.5%	189
June	162,159	93.8%	0
July	178,007	99.7%	**28,544**
August	178,534	100.0%	495
September	129,509	**74.9%**	138
October	88,304	**49.5%**	37
November	170,683	98.8%	656
December	176,579	98.9%	166

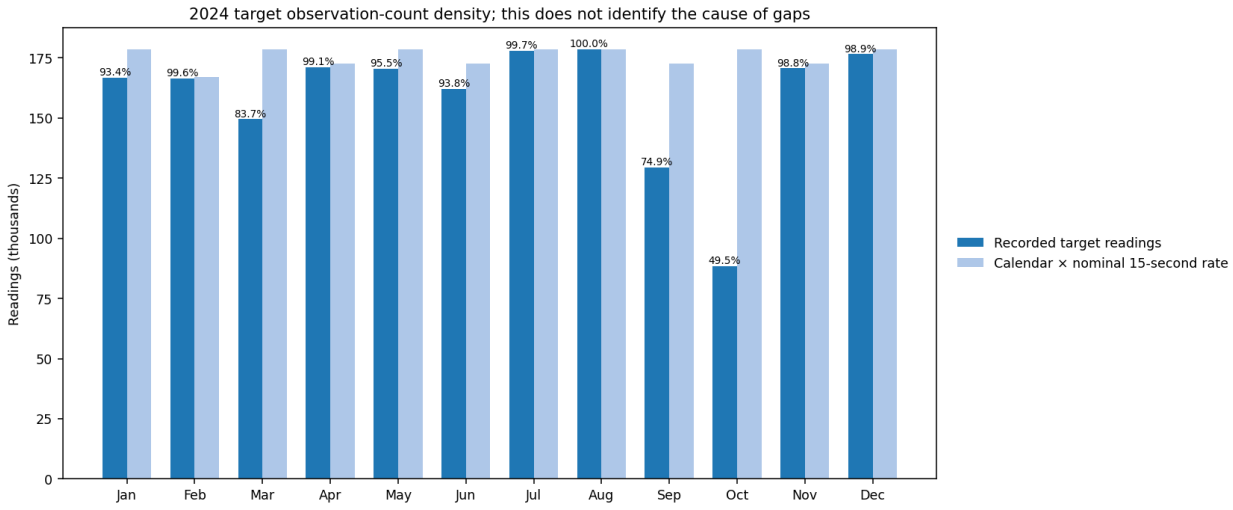


Figure 1: Monthly target observation-count density relative to a nominal 15-second calendar schedule. The chart does not diagnose the reason for missing observations.

October shaft-power readings appear on **16 of 31 calendar dates**. Lower density in September and October makes coverage-stratified evaluation necessary, without assuming a particular cause of missing records.

3. Sensor quality and candidate measurements

The 36 observed 2024 channels consist of one target, six torquemeter-related channels or close proxies, four draft channels with constant zero readings, one water-depth channel without October target overlap, and **24 initially eligible predictors**. The predictors comprise **four other-source engine**, **four vessel-motion**, **two onboard relative-wind**, and **14 environmental provider** channels.

The four draft sensors have **8,226,522 values across the thirteen files, all equal to 0.0 m**. They do not provide useful draft variation. This observation does not establish whether the source is installation, reporting or

export behavior. The eastward and northward sea-current variables are assigned to their corresponding provider sources; velocity is expressed in **m/s**. Sea-water temperature is in degrees Celsius, and the registration does not specify a unit for shaft revolutions.

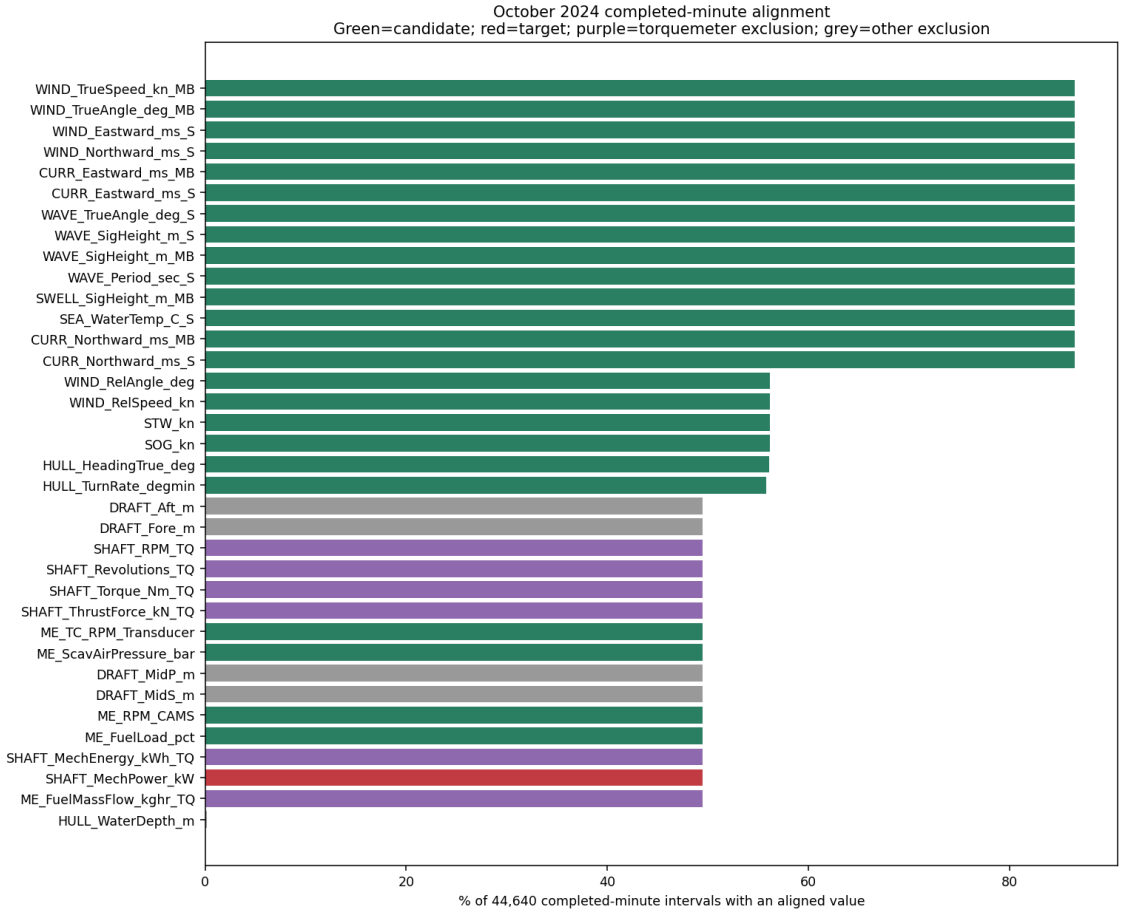


Figure 2: Share of October calendar minutes represented after alignment, grouped by whether a channel is a candidate input or excluded measurement.

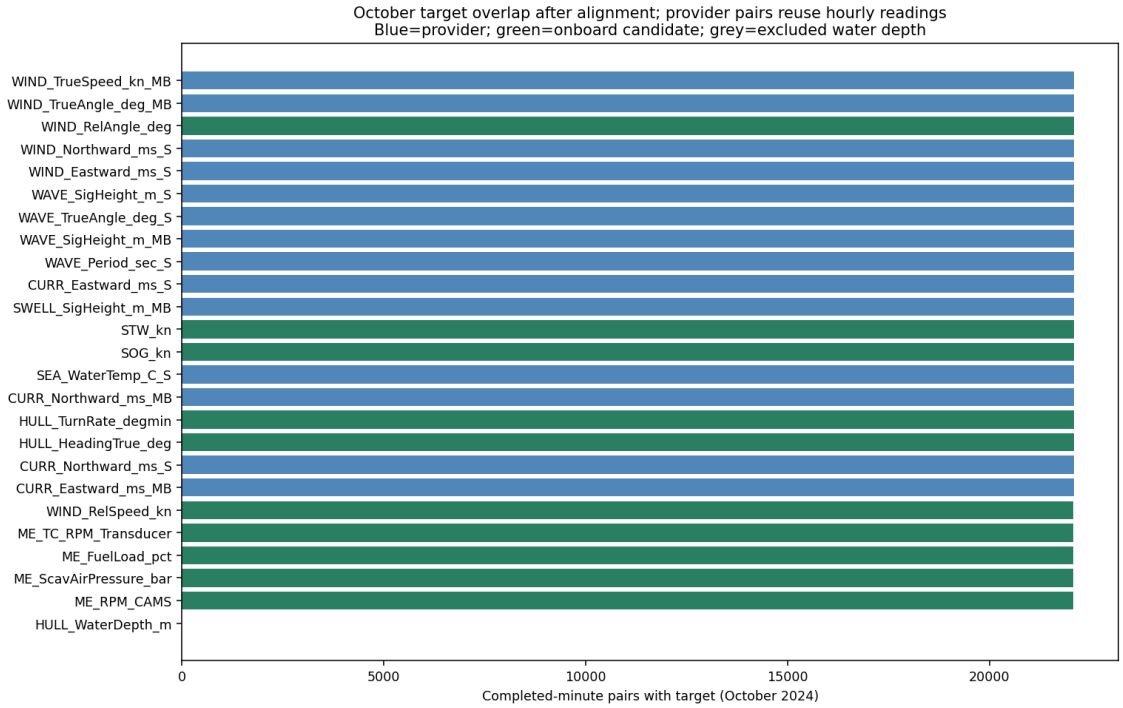


Figure 3: Number of October completed-minute intervals in which each channel is paired with the shaft-power target. Multiple minute pairs may share a single provider measurement.

3.1 Target-power anomalies

July 2024. The shaft-power series contains **28,544 negative readings** across fourteen dates. All target readings on 11, 12 and 15 July are negative. The longest near-contiguous negative sequence spans **17,399 readings**. At completed-minute resolution, **6,948 negative-power intervals** also coincide with positive CAMS engine RPM and speed through water above 3 knots.

November 2024. There are **656 negative readings** across eight dates, with a minimum near **24,757 kW**. The combined negative-power/positive-engine-operation condition occurs in **93 one-minute intervals**.

These patterns support a **target-quality concern** and separate assessment of July and November. They do not on their own distinguish calibration errors from reversal, sign convention or other operating circumstances. Negative observations have not been silently clipped or replaced. June has no negative target readings, but that absence alone is not evidence of clipping.

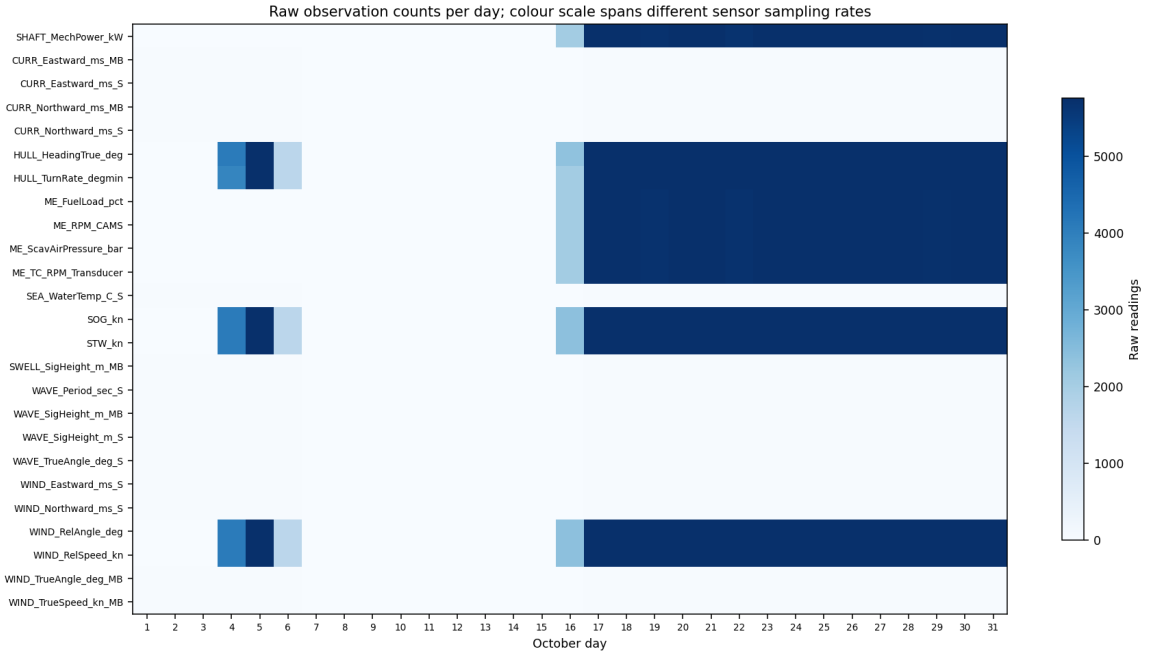


Figure 4: Daily October raw observation counts across sensor channels. Colours reflect different source sampling rates and should not be compared as equivalent uptime percentages.

4. Relationships observed in October

Pairwise Pearson correlations describe the aligned October minutes for scalar candidate variables. Torquemeter-related features, constant draft, sparse water depth and circular directions were excluded from this correlation display.

Measurement	Pearson correlation with shaft power	Paired minutes
Scavenging-air pressure	+0.978	22,087
Turbocharger RPM (transducer)	+0.950	22,087
Engine fuel load (%)	+0.920	22,087
Engine RPM (CAMS)	+0.880	22,086
Speed over ground	+0.638	22,088
Speed through water	+0.628	22,088
Significant wave height (Provider MB)	+0.256	22,088

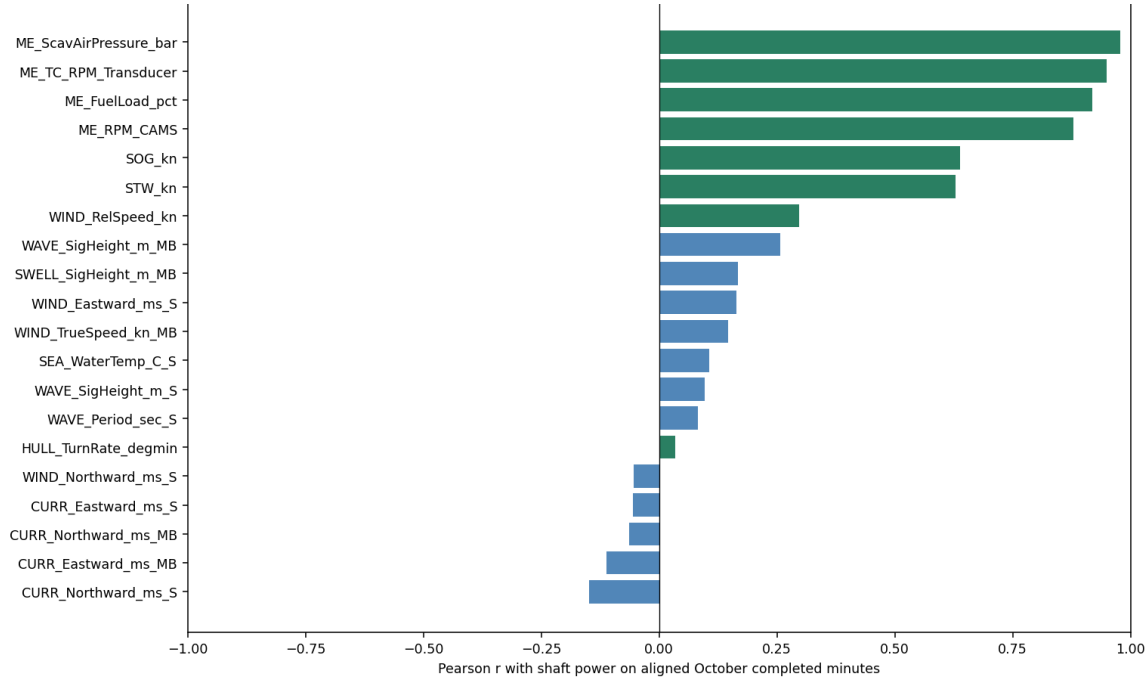


Figure 5: Descriptive Pearson correlations between shaft power and selected aligned scalar measurements. Blue bars indicate external sources for which repeated hourly values contribute to many minute pairs.

The strong engine associations indicate that these signals may be informative for reconstructing shaft power, subject to availability and independence checks. The weather correlations use repeated provider values and serially related target minutes; **22,088 paired minutes are not equivalent to 22,088 independent weather observations**. These associations are not causal effects or trained-model feature importances.

The October speed–power relationship is also not consistently increasing. Among nonnegative speed and power intervals, the observed median decreases from approximately **2,997 kW** at 7.0–7.5 knots to **2,564 kW** at 10.0–10.5 knots and **2,431 kW** at 11.0–11.5 knots. An unconstrained descriptive cubic fit likewise falls from approximately 2,955 kW at 7 knots to 2,576 kW at 10 knots. In contrast, the familiar idealized **power speed³** expression is a physical heuristic under assumptions that these pooled observations do not satisfy. Loading, operating conditions, time effects and measurement quality have not been controlled sufficiently to explain the empirical decline.

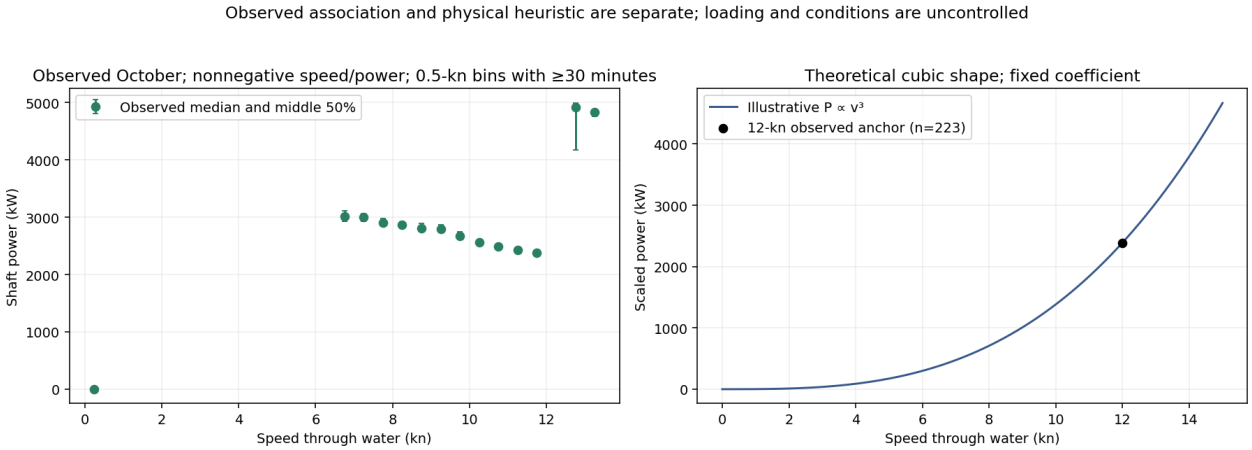


Figure 6: Observed October speed–power summaries and an illustrative cubic physical heuristic, shown separately. The theoretical shape is not an empirical fit for this vessel.

5. Noon Reports and implications for modeling

The twelve Noon Report datasets cover **all 366 unique dates in leap year 2024**. Their daily cadence is much coarser than sensor records. June includes missing fuel type and propulsion-consumption fields; **18 extreme slip entries** were flagged during the year, including 60% and +50% values. Daily displacement could act as a loading indicator, but a same-day measurement may not have been available early on that date. Incorporating it requires documented measurement and issuance times or a defensible lag.

The provisional chronological design is **January–June training, August validation and December final testing**. July and November are reserved for quality investigation, while September–October provide coverage-stress analysis. December's **166 negative target readings** require an evaluation policy established before inspecting final test errors.

6. Interpretation and limitations

The current findings support a coherent starting point for regression, but they do not establish predictive accuracy, fuel savings or deployment readiness. The main dependencies are the torquemeter's trustworthy operating periods, provider timestamp semantics, engine-signal availability and the generalizability of relationships across voyages and loading conditions. A completed-minute reconstruction should not be described as a real-time forecast without further evidence.